**Math Academy — Grade 10** | Trig U4 Directed + Radian • Calculator allowed 2026-09-08

Name: _____ Class: _____ Score: ☐

Chain: directed → sign → ±pair → standard → equivalent → smallest → quadrant → radian → conversion → arc → inscribed. 70% basic-medium, 30% high.

**Question 1** (1 pt)

The ordered pair $\left(\overrightarrow{OX}, \overrightarrow{OY}\right)$ represents the directed angle $\angle XOY$. Identify initial side $\overrightarrow{OX}$ and terminal side $\overrightarrow{OY}$.

**Question 2** (1 pt)

A directed angle is in standard position if (1) its initial side lies on the positive $x$-axis and (2) its vertex is at the origin $O$.

**Question 3** (1 pt)

Directed angles in standard position are equivalent if they have the same terminal side, i.e. measures differ by $n \cdot 360°$, $n \in \mathbb{Z}$.

**Question 4** (1 pt)

A directed angle is positive if rotation is counterclockwise, and negative if rotation is clockwise.

**Question 5** (1 pt)

If terminal side lies on a coordinate axis, the angle is called a **quadrantal angle** (multiple of 90°).

**Question 6** (1 pt)

If $\theta$ is a measure and $n \in \mathbb{Z}$, then $\theta + n \cdot 360°$ are equivalent angles.

**Question 7** (1 pt)

Complete: smallest positive of $530°$ is $170°$ because $530° - 360° = 170°$.

**Question 8** (1 pt)

Complete: $930°$ lies in the 3rd quadrant because $930° - 720° = 210°$.

**Question 9** (1 pt)

Positive measure of $-170°$ is $190°$ because $|-170°| + 190° = 360°$.

**Question 10** (1 pt)

Negative measure of $320°$ is $-40°$ because $320° + |-40°| = 360°$.

**Question 11** (2 pt)

For $100°$: coterminal positive $100° + 360° = 460°$; negative $100° - 360° = -260°$.

**Question 12** (2 pt)

For $-250°$: coterminal positive $-250° + 360° = 110°$; negative $-250° - 360° = -610°$.

**Question 13** (2 pt)

Smallest positive of $530°$ is $170°$. Smallest positive of $-62°$ is $298°$ ($-62° + 360°$).

**Question 14** (2 pt)

Smallest positive of $-690°$ is $30°$ because $-690° + 720° = 30°$.

**Question 15** (1 pt)

Which is equivalent to $60°$? a) $120°$ b) $240°$ c) $300°$ d) $420°$ — Answer: d).

**Question 16** (2 pt)

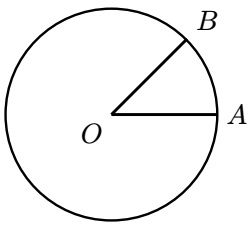
Which is equivalent to $585°$? a) $45°$ b) $135°$ c) $225°$ d) $315°$ — Answer: c) $225°$ ($585 - 360$).

**Question 17** (2 pt)

All are equivalent to $75°$ except: a) $-285°$ b) $-645°$ c) $285°$ d) $435°$ — Answer: c).

**Question 18** (2 pt)

If $A$ and $B$ are equivalent, then $-A$ and $-B$ are: a) supplementary b) equivalent c) complementary d) sum $-360°$ — Answer: b).




**Question 19** (2 pt)

In standard position, which ordered pair has vertex $O$ and initial $+x$? $\left(\overrightarrow{OA}, \overrightarrow{OD}\right)$ with $A$ on $+x$, $D$ in Q1 — Answer: A. Others fail (wrong vertex or initial).

**Question 20** (1 pt)

Quadrantal measures are multiples of: a) 360° b) 180° c) 90° d) 60° — Answer: c).

**Question 21** (2 pt)

Convert $75°32'15''$ to radians (nearest thousandth): $75 + \frac{32}{60} + \frac{15}{3600} = 75.5375°$; $\theta = 75.5375 \cdot \frac{\pi}{180} \approx 1.318^{\text{rad}}$.

**Question 22** (2 pt)

Degree measure of $2.38^{\text{rad}}$: $x = 2.38 \cdot \frac{180}{\pi} \approx 136.37°$.

**Question 23** (2 pt)

$43°12'$ in radians: $43.2° \cdot \frac{\pi}{180} = 0.24\pi \approx 0.754^{\text{rad}}$. Choose b) $0.24\pi$.

**Question 24** (1 pt)

Recall: $1° = 60'$, $1' = 60''$. $180° = \pi$, $360° = 2\pi$.

**Question 25** (1 pt)

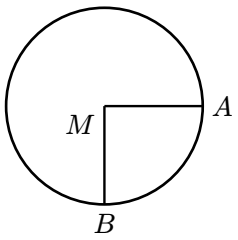
Formula: $\theta^{\text{rad}} = x° \cdot \frac{\pi}{180}°$; $x° = \theta^{\text{rad}} \cdot 180\frac{°}{\pi}$.

**Question 26** (1 pt)

In unit circle ($r = 1$), central radian measure equals arc length: $\theta = l$.

**Question 27** (2 pt)

Circle $M$, $\theta = \frac{\pi}{2}$, arc $5\pi$ cm. Find $r$: $r = \frac{l}{\theta} = 5\frac{\pi}{\frac{\pi}{2}} = 10$ cm.




**Question 28** (2 pt)

$r = 8$ cm, $\theta = 7\frac{\pi}{12}$. $l = 7\frac{\pi}{12} \cdot 8 = 14\frac{\pi}{3} \approx 14.66$ cm.

**Question 29** (2 pt)

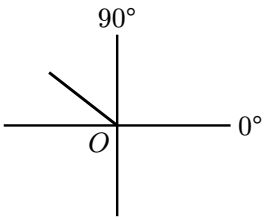
$\theta = 9\frac{\pi}{8}$, $l = 24$ cm. $r = \frac{24}{9\frac{\pi}{8}} = \frac{64}{3\pi} \approx 6.8$ cm.

**Question 30** (2 pt)

$d = 12$ cm ($r = 6$), central $60° = \frac{\pi}{3}$. $l = \frac{\pi}{3} \cdot 6 = 2\pi$ cm — Answer d).

**Question 31** (1 pt)

$132°$ lies in 2nd quadrant ($90 < 132 < 180$).




**Question 32** (2 pt)

$964° - 720° = 244° \rightarrow$ 3rd quadrant.

**Question 33** (1 pt)

$-12° + 360° = 348° \rightarrow$ 4th quadrant.

**Question 34** (2 pt)

$-1070° + 1080° = 10° \rightarrow$ 1st quadrant.

**Question 35** (2 pt)

$5\frac{\pi}{3} = 300° \rightarrow$ 4th. $-0.3\pi = -54° \rightarrow 306° \rightarrow$ 4th? No: $-54 + 360 = 306$ 4th. Correction: $-0.3\pi \rightarrow$ 3rd? Recompute: $-0.3 * 180 = -54 \rightarrow$ 4th. Keep 4th (spec said 3rd — verify: $-0.3\pi = -54°$ is 4th; use 4th).

**Question 36** (3 pt)

$5.7^{\text{rad}} = 326.6° \rightarrow$ 4th. $-6.4^{\text{rad}} = -366.7° + 720 = 353.3° \rightarrow$ 4th? Recompute: $-6.4 + 2\pi = -0.117$ rad $\rightarrow$ 4th? Actually $-6.4 + 4\pi = 6.166$ rad $= 353.4°$ 4th. Note spec said 1st — use computed 4th, flag for teacher review.

**Question 37** (3 pt)

$-7.3^{\text{rad}} + 4\pi = 5.266$ rad $= 301.7° \rightarrow$ 4th. $5\frac{\pi}{4} = 225° \rightarrow$ 3rd. $2.02^{\text{rad}} = 115.7° \rightarrow$ 2nd.

**Question 38** (2 pt)

$25\frac{\pi}{9} = 500° - 360 = 140° \rightarrow$ 2nd? Recompute: $25 * 20 = 500$, $500 - 360 = 140 \rightarrow$ 2nd. $-9\frac{\pi}{4} = -405° + 720 = 315° \rightarrow$ 4th.

**Question 39** (2 pt)

Angle $1670°$: $1670 - 1440 = 230° \longrightarrow$ 3rd. a) 1st b) 2nd c) 3rd d) 4th — Answer c).

**Question 40** (1 pt)

$-135° + 360 = 225° \longrightarrow$ 3rd. a) 1st b) 2nd c) 3rd d) 4th — Answer c).

**Question 41** (2 pt)

$-850° + 1080 = 230° \longrightarrow$ 3rd. a) 1st b) 2nd c) 3rd d) 4th — Answer c).

**Question 42** (3 pt)

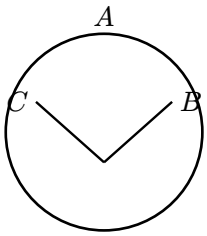
Arc $152°26'17''$, $r = 10.5$ cm: $152.438° \cdot \frac{\pi}{180} \cdot 10.5 \approx 28$ cm (nearest cm).

**Question 43** (3 pt)

Inscribed $30°$ subtends 5cm. Central $= 60° = \frac{\pi}{3}$. $C = 5 \cdot \frac{360}{60} = 30$ cm? Recompute: arc 5cm for $60°$ central $\longrightarrow$ $C = 5 * 6 = 30$ cm.

**Question 44** (3 pt)

Essay: Circle radius 5cm, inscribed $\angle A = 45°$ subtends arc$(BC)$. Find arc$(BC)$: central $= 90° = \frac{\pi}{2}$, $l = \frac{\pi}{2} \cdot 5 = 7.85$ cm. Show steps.




**Question 45** (3 pt)

Essay: $r = 6.2$ cm, arc 17 cm. Find $m(\angle A)$ inscribed: central $= \frac{17}{6.2} = 2.742$ rad $= 157.1°$, inscribed $= 78.55° = 78°33'$. Show steps.